

SHP ASSET MANAGER

CABLE TRIGGER

User Manual



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HARDWARE OVERVIEW

The **Cable Trigger** is a servo-driven shutter release that physically actuates your camera's cable-release port. A small servo arm presses the plunger to a position you dial in through the app – no radio frequency tricks, no electronic handshakes – just a mechanical press of the cable release your camera already knows and trusts.

Because it works at the mechanical level it is compatible with cameras that do not expose an electronic remote-trigger port, and it produces the same signal your camera's built-in shutter button produces.

WHAT THE APP CONTROLS

PARAMETER	WHAT IT SETS
Pre-Trigger Delay	Time between the app sending the trigger command and the servo moving. Use to give yourself time to step away before the shot fires, or to synchronise with another event.
Exposure Time	How long the servo holds the cable release depressed. Match this to your camera's bulb-exposure requirement, or the minimum dwell time the shutter needs to register a press.
Trigger Engagement Amount	The servo angle (expressed as a percentage of its full range) used during each trigger. Each camera and cable-release combination has a different sweet spot – this dials it in precisely.

CONNECTIVITY

The Cable Trigger communicates with the SHP Asset Manager app over **Bluetooth Low Energy (BLE)** and over a **WiFi hotspot** it creates itself. Both modes provide full, real-time control of all settings.

SYSTEM STATUS

The large status display at the top of the Cable Trigger tab shows the current state of the device and a COUNTDOWN timer. States you will see:

STATE	WHAT THE DEVICE IS DOING
IDLE	Ready and waiting. Tap TRIGGER SEQUENCE to fire.
DELAYING	Pre-trigger delay is counting down before the servo moves.

STATE	WHAT THE DEVICE IS DOING
MOVING OUT	Servo is travelling to the engagement position.
HOLDING	Servo is holding the cable release depressed for the Exposure Time duration.
RETURNING	Servo is returning to the home position after the exposure.

The arc gauge below the status display animates in real time, sweeping from 0% (home) to the set engagement amount as the servo travels.

STATUS LIGHT

The Cable Trigger has a small status LED on the board. It tells you at a glance what the device is doing – useful the moment you power on and before the app has connected.

LED PATTERN	WHAT IT MEANS
Two quick flashes at boot	Firmware is running – the board powered up correctly.
Slow pulse (brief flash every ~2 s)	Advertising over BLE – waiting for the app to connect.
Steady on	Connected and idle – ready to trigger.
Fast blink (on/off every 125 ms)	Trigger active – servo is moving or holding the cable release.

If the LED is off entirely, the device is in deep sleep. Wake it by pressing its power or wake button, or cycling power.

CONNECTING TO THE APP

The SHP Asset Manager app controls all Cable Trigger settings in real time. Connect via **Bluetooth** or **WiFi** – both give full control. WiFi is recommended for firmware updates due to significantly faster transfer speeds.

BLUETOOTH

- 1 Power on the Cable Trigger.
- 2 Ensure Bluetooth is enabled on your phone.
- 3 Open the app and tap the **CABLE TRIGGER** tab at the bottom of the screen.
- 4 Tap ⚡ **CONNECT DEVICE** in the header and select your device from the scan list.
- 5 Settings sync automatically. The dot in the header turns orange when connected.

The app remembers your device. On subsequent launches it reconnects silently if the Cable Trigger is in range. Tapping **CONNECT DEVICE** when a device is already saved gives you the option to reconnect to it, scan for a new device, or connect via WiFi.

WIFI

The Cable Trigger creates its own WiFi hotspot. WiFi is recommended when installing firmware updates – it is significantly faster than Bluetooth.

- 1 Power on the Cable Trigger.
- 2 On your phone, join the Cable Trigger's network in your OS WiFi settings. If you have set a Security PIN of 8 or more characters in the app, that PIN is also the hotspot password.
- 3 Return to the app. Tap ⚡ **CONNECT DEVICE** → *Connect via WiFi*.
- 4 Enter IP **192.168.4.1** and port **80**, then tap **CONNECT**.
- 5 Settings sync automatically once connected.

PI TX CONTROL

If you use a **Pi TX** (Raspberry Pi transmitter), the Cable Trigger can also be controlled from the Pi TX browser panel. For local field use, join the secured WiFi network whose name begins with `PI_TX_` and open `http://10.42.0.1`; no internet is required. If the Pi is on a trusted upstream LAN, `http://pi-tx.local` or its LAN IP can be used instead. The Pi TX browser panel mirrors the app's Cable Trigger controls, so you can trigger, adjust

settings, and run a longevity test without the phone present. See the [Lazer Eye manual](#) for Pi TX setup details.

SETTINGS

Tap the  **gear icon** in the header to open Settings. From here you can check for firmware updates, change the device name, set a security PIN, and adjust app preferences.

CONFIGURATION SETTINGS

The **CONFIGURATION** card is the main control panel for the Cable Trigger. Tap its header to expand or collapse it. All values are written to the device immediately when changed and are retained after power-off.

PRE-TRIGGER DELAY

0 s to 18,000 s

Time between the trigger command leaving the app and the servo starting to move. Increase to give yourself time to step back from the camera. Small values step in 1-second increments; above 60 s they step in 1-minute increments; above 1 hour they step in 10-minute increments.

EXPOSURE TIME

0 s to 18,000 s

How long the servo holds the cable release depressed. For a standard shot, set this slightly longer than your camera's minimum registration time. For bulb exposures, set it to the full exposure length. Follows the same step increments as Pre-Trigger Delay. Setting this to 0 still sends a brief minimum pulse to the firmware so the servo completes a full actuation cycle.

The compact summary line at the top of the collapsed CONFIGURATION card shows current values at a glance: DELAY · EXP · ENG.

TRIGGERING

Tap the **TRIGGER SEQUENCE** button at the bottom of the status card to fire the shutter immediately. The button is only active when the device is IDLE and connected (or in simulation mode).

TRIGGER ENGAGEMENT AMOUNT

Every cable release has a pre-travel zone – a short distance the plunger moves before it actually trips the shutter. **Trigger Engagement Amount** sets how far the servo travels (as a percentage of its full range) to engage the release. This is the single control that determines servo rotation depth.

Tap the **TRIGGER ENGAGEMENT AMOUNT** row inside the CONFIGURATION card to expand it. A

`ParamControl` slider with +/- buttons lets you set the value from **1% to 100%**.

The arc gauge in the status card animates in real time as you adjust the value, giving live visual feedback of where the servo will travel to.

FINDING THE RIGHT ENGAGEMENT AMOUNT

- 1 Attach the cable release to your camera and mount the Cable Trigger.
- 2 Set Pre-Trigger Delay and Exposure Time to a comfortable value).
to (e.g.).
- 3 Start with a low Trigger Engagement Amount (e.g. 20%) and tap **TRIGGER SEQUENCE** .
- 4 Gradually increase the percentage until the shutter fires reliably on every press.
- 5 Find the minimum % that fires reliably – this avoids over-stressing the release mechanism.

Do not set Trigger Engagement Amount higher than needed. Over-driving a cable release can stress the mechanism and damage the camera's shutter port.

All settings are written to the device immediately when changed and survive power cycles.

PRESETS

Presets save a complete trigger configuration – Pre-Trigger Delay, Exposure Time, and Trigger Engagement Amount – under a name you choose. Up to **five presets** can be stored. They are saved on your phone and are not affected by device connections or firmware updates.

The PRESETS row appears at the top of the expanded CONFIGURATION card.

SAVING A PRESET

- 1 Dial in your settings (Pre-Trigger Delay, Exposure Time, Trigger Engagement Amount) for a particular camera or shooting scenario.
- 2 In the PRESETS row, tap **SAVE CURRENT AS** button, **+** button if presets the dashed **PRESET** or the **SAVE** already exist.
- 3 Type a name (up to 16 characters) and tap the **✓** confirm button or press Return.

LOADING A PRESET

Tap a preset chip in the horizontal PRESETS list to load it instantly. All three values (Pre-Trigger Delay, Exposure Time, Trigger Engagement Amount) are applied immediately and synced to the device if connected.

DELETING A PRESET

Tap the **×** button on the right side of any preset chip to delete it. The slot is freed and a new one can be saved in its place.

Each preset chip shows a compact summary: **D:** = Pre-Trigger Delay, **E:** = Exposure Time, **R:** = Trigger Engagement Amount %.

SHUTTER CALIBRATION

Shutter Calibration uses your phone's microphone to listen to the actual shutter click and measure exactly when it opens and closes. It calculates the offset between what the firmware held and how long the shutter was actually open, so you can apply a correction to Exposure Time for your specific camera body.

The SHUTTER CALIBRATION card is a separate collapsible section below the CONFIGURATION card. Tap its header to expand it.

RUNNING A CALIBRATION TEST

- 1 Dial in Trigger Engagement Amount first – the test fires a real shot, so the servo must already be reaching the shutter.
- 2 Enter a name for this calibration (e.g. your **LENS /** field at the top of camera body or lens) in the **SHUTTER NAME** the section.
- 3 Hold your phone close to the camera (within 15-20 cm), microphone facing the camera body where the shutter sound is loudest.
- 4 Make sure the area is quiet – ambient noise reduces detection accuracy.
- 5 Tap **TEST** . The app counts down, fires one shot, and records audio during **SHUTTER** the cycle.
- 6 After a brief analysis phase, the EXPECTED / DETECTED / OFFSET metrics appear, with a waveform showing the open and close click peaks.
- 7 Review the offset. A positive offset means the firmware held longer than the shutter was open; a negative offset means it closed before the hold ended.
- 8 Tap **SAVE PROFILE** to store the result under the name you entered.

APPLYING A SAVED PROFILE

Saved profiles appear in the **SAVED PROFILES** list at the bottom of the Shutter Calibration section. Each row shows the profile name, the date it was tested, and its offset in milliseconds.

Tap **APPLY** on any profile to adjust the firmware's Exposure Time for the current session. The app calculates the corrected hold time (current Exposure Time + offset) and sends it to the device. Confirm in the dialog that appears.

Applied calibration lasts for the current connection session. If you disconnect and reconnect, the original Exposure Time is restored. Apply the profile again each time you connect if your shooting consistently requires the correction.

Grant microphone permission when prompted. Without it the app cannot record audio and TEST SHUTTER will fail with a permission error.

WHEN TO RECALIBRATE

- > When you switch to a different camera body.
- > When you change the cable release mechanism.
- > When you find the shutter is consistently held too short or too long after a firmware update.
- > When shooting in new temperature conditions (thermal expansion can shift the actuation point slightly).

REMOTE TRIGGER MODE

Remote Trigger Mode lets you fire the camera without tapping the on-screen TRIGGER SEQUENCE button. The toggle is at the bottom of the CONFIGURATION card, below the Longevity Test row.

The screen stays on automatically while Remote Trigger Mode is active, so the phone does not sleep between shots. All shot settings (Pre-Trigger Delay, Exposure Time, Trigger Engagement Amount) still apply when firing remotely.

ANDROID — VOLUME BUTTONS

While Remote Trigger Mode is on, pressing the physical **Volume Up** or **Volume Down** buttons on your Android phone fires the shutter. This lets you hold the phone in one hand and trigger with a thumb, keeping your other hand free to steady the camera.

IOS — BLUETOOTH CLICKER

On iOS, pair a Bluetooth shutter remote (a BLE clicker, such as the kind sold for selfies or remote presentations) with your phone. While Remote Trigger Mode is on, pressing the clicker fires the Cable Trigger. The clicker can be held in one hand or placed on a surface nearby.

PLATFORM	REMOTE INPUT	HOW TO ACTIVATE
Android	Volume Up or Volume Down (hardware buttons)	Enable the Remote Trigger Mode toggle inside the CONFIGURATION card. No additional hardware required.
iOS	BLE shutter clicker (paired in iOS Bluetooth settings)	Pair the clicker via iOS Settings → Bluetooth, then enable the Remote Trigger Mode toggle in the app.

Disable Remote Trigger Mode when you're done shooting to restore normal volume-button behaviour on Android.

LONGEVITY TEST

The **LONGEVITY TEST** runs the Cable Trigger through a set number of actuation cycles autonomously, using your current Pre-Trigger Delay, Exposure Time, and Trigger Engagement Amount settings. Use it to burn in a new cable release mechanism, verify reliability under repeated use, or simulate a long shoot.

Expand the **LONGEVITY TEST** row inside the **CONFIGURATION** card.

RUNNING A LONGEVITY TEST

- 1 Set the **CYCLES** using the +/- buttons or type directly into the counter field number of (1 to 9999 cycles).
- 2 Tap **START** . The device begins firing immediately, repeating the full trigger **TEST** sequence for the requested number of cycles.
- 3 The progress display shows cycles remaining. The test continues even if you disconnect the phone.
- 4 Tap **STOP TEST** at any time to abort.

The longevity test runs autonomously on the device – it is safe to disconnect the phone once it has started. The test uses your current delay, exposure, and engagement settings, so dial those in before starting.

LINKING TO A LAZER EYE

The Cable Trigger can be triggered automatically by a **Lazer Eye** LiDAR sensor in two ways. Both are configured from the **LAZER EYE tab** of the SHP Asset Manager app.

DIRECT ESP-NOW CROSS-LINK

The Lazer Eye sends a wireless trigger packet directly to the Cable Trigger over **ESP-NOW** every time the LiDAR detects a subject. Once paired, this link operates autonomously – no phone needs to be connected after setup. The Cable Trigger fires using whatever Pre-Trigger Delay, Exposure Time, and Trigger Engagement Amount settings are currently stored on it.

Lazer Eye firmware **v2.9.4 or later** is required for confirmed-delivery status. Older firmware sends the packet but cannot confirm the Cable Trigger received it.

- 1 Connect **Cable** to the app via BLE (the orange dot in the Cable Trigger tab the **Trigger** header confirms the connection).
- 2 Open the **LAZER EYE tab** and connect the Lazer Eye (WiFi, BLE, or Pi TX).
- 3 Expand **DIRECT CABLE** section. **GET** – the app reads the Cable Trigger's ESP radio the **TRIGGER LINK** Tap **MAC** MAC address over BLE and displays it.
- 4 Tap **PAIR** . The Lazer Eye stores the Cable Trigger's MAC address and begins firing it on every accepted LiDAR detection.
- 5 Tap **TEST** to send a test packet. Listen for a shutter click from the Cable Trigger. **LINK** If delivery is confirmed you will see a green "Delivery confirmed" message; on older Lazer Eye firmware you will see a yellow "Sent (delivery not confirmed)" message.

Both devices remember the pairing after power cycling – you only need to pair once. To remove the link, expand **DIRECT CABLE TRIGGER LINK** and tap **UNPAIR**.

BLE-MEDIATED LINK

While both the Cable Trigger and Lazer Eye are connected to the app simultaneously, you can enable the **FIRE CABLE TRIGGER ON LIDAR EVENT** toggle in the **CABLE TRIGGER LINK** section of the Lazer Eye tab. When enabled, the app relays each accepted LiDAR trigger to the Cable Trigger over BLE in real time.

- > Each relay fires using the Cable Trigger's own **shots-per-trigger** setting.
- > If the Cable Trigger is mid-sequence when an event arrives, that event is **skipped** – events are not queued.

- > The link stops automatically if either device disconnects. No pairing is stored; the toggle must be re-enabled each session.

CHOOSING THE RIGHT APPROACH

SCENARIO	RECOMMENDED LINK
Unattended setup – phone pocketed or left behind	Direct ESP-NOW Cross-Link – operates without a phone after pairing.
Phone already monitoring both devices simultaneously	BLE-Mediated Link – no extra pairing required; toggle on when needed.
Maximum reliability & confirmed delivery	Direct ESP-NOW Cross-Link with Lazer Eye firmware v2.9.4+.

TROUBLESHOOTING

Camera doesn't fire when TRIGGER SEQUENCE is tapped

Increase **Trigger Engagement Amount** – the servo may not be travelling far enough to press the plunger. Increase gradually and test until the shutter fires reliably. Also confirm the cable release is securely attached to the camera port.

Camera fires inconsistently

Increase **Exposure Time** – the servo may not be pressing the release long enough for the camera to register the signal. Run **Shutter Calibration** (TEST SHUTTER) to measure the gap and apply a correction automatically.

Servo strains or over-drives the cable release

Lower **Trigger Engagement Amount**. Find the minimum percentage that reliably fires the shutter – this reduces mechanical stress on the release and the camera's shutter port.

TEST SHUTTER shows no detection result or an error

Hold your phone closer to the camera (within 15 cm) with the microphone facing the shutter. Ensure the room is quiet. Check that microphone permission is granted for the SHP Asset Manager app in Settings → Privacy → Microphone. Also confirm Trigger Engagement Amount is already set high enough that the shutter actually fires during the test.

Applying a calibration profile has no visible effect

The APPLY action sends a corrected Exposure Time for the current session only. Disconnect and reconnect to restore the original value, then APPLY again. If the correction is consistently needed, manually adjust **Exposure Time** in the CONFIGURATION card to match the corrected value.

Volume buttons don't trigger on Android

Confirm **Remote Trigger Mode** is toggled on (label turns orange and the toggle is highlighted). The app must be in the foreground. If another app has captured the hardware button events, close it and try again.

BLE clicker doesn't fire on iOS

Confirm the clicker is paired in iOS Settings → Bluetooth and shows as connected. Then confirm **Remote Trigger Mode** is on in the app. The app must be in the foreground to receive clicker events.

Can't connect via Bluetooth

Toggle Bluetooth off and on and power-cycle the Cable Trigger. On iOS: Settings → Privacy → Bluetooth → confirm

the SHP Asset Manager app has permission. The Cable Trigger appears in the scan list by the name set in Hardware Settings (the default name is shown on the device label).

Settings revert after reconnecting

Settings are written to the device immediately while connected. Ensure the device is connected (orange dot in header) when you make changes. The Shutter Calibration APPLY correction is session-only and must be re-applied each connection if needed permanently.

A firmware update is slow or failing

Switch to a WiFi connection – updates over WiFi are dramatically faster than Bluetooth. Open Settings → Hardware Settings → CHECK FOR UPDATES, then connect via WiFi first if needed.

Direct link shows "sent (delivery not confirmed)" instead of "delivery confirmed"

Confirmed delivery requires **Lazer Eye firmware v2.9.4 or later**. Update the Lazer Eye via the Lazer Eye tab → Settings → CHECK FOR UPDATES. After updating, re-pair (UNPAIR then repeat the GET MAC → PAIR steps) and TEST LINK again.

Direct ESP-NOW link set up but Cable Trigger doesn't fire

Confirm ESP-NOW is enabled on the Lazer Eye (check the ESP-NOW section of the Lazer Eye tab). Keep both devices within range during initial testing. If TEST LINK shows an error, UNPAIR and redo the pairing – the Cable Trigger BLE connection must be active during GET MAC. After re-pairing, range of up to ~100 m line-of-sight is typical.

BLE-mediated link fires inconsistently or skips events

Both the Cable Trigger and Lazer Eye must remain connected to the app throughout the session – the link stops if either disconnects. Events are **not queued**: if the Cable Trigger is mid-sequence when a LiDAR event arrives, that event is skipped. Reduce your LiDAR sensitivity or increase inter-detection interval to lower the event rate relative to the Cable Trigger's cycle time.

Direct link stops working after powering both devices off and back on

Pairing is retained in flash memory on both devices and should survive power cycles. If the link stops: open the Lazer Eye tab, expand DIRECT CABLE TRIGGER LINK, and confirm it still shows PAIRED. If it shows NOT PAIRED, the Lazer Eye flash was cleared (e.g. by a firmware reflash) – redo the GET MAC → PAIR steps. Also confirm ESP-NOW is still enabled on the Lazer Eye.

UPDATING FIRMWARE

- 1 Connect to the Cable Trigger in the app.
- 2 Tap the  **gear icon** in the header to open Settings.
- 3 Scroll to **Hardware Settings** and tap **CHECK FOR UPDATES** .
- 4 If an update is available, tap **INSTALL UPDATE** . The device reboots and the app reconnects automatically.

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[Hardware Build Guide](#) · [Firmware & Software Install](#) · [Lazer Eye Manual](#)
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